

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 219 — Past Questions & Materials
by Renance

Course: MTH 219
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 200

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Paper 1 of 2 — MTH 219(2018 AND 2019)

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
B.Sc. (MATHEMATICS) DEGREE FIRST SEMESTER EXAMINATION 2018/2019
COLLEGE OF SCIENCE, DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE
TIME: 2 HOURS

MTH 219- INTRODUCTION TO OPERATION RESEARCH

INSTRUCTION: ANSWER QUESTION ONE AND ANY OTHER THREE QUESTIONS

1. (a) Using the graphical method needed for model X and Y of

Maximize $Z = 80x + 120y$, for Model Y. for each of X and

Subject to the constraints Y, kg of r, is required. It takes 2

hours to manufacture Model X

and 2 hours to manufacture Model

Y. How many units of each Model

should be produced to maximize

$20x + 5y$ the profit using the graphical

method?

(b) Minimize $Z = 20x + 10y$

Subject to the constraints 3. (a) Write a short note on the

following 7

3x+y (i) Feasible solution

4x+3y (ii) — Infeasible solution

xy (iii) Feasible region

(c) Use the simplex method to solve (iv) Optimal solution ;

the linear programming problem (v) Corner point feasible

solution

Maximize $Z = 3x + 5y + 4k$

Subject to the constraints (b) Convert the following primal

linear programming model problems

$2x + 3y < 8$ to dual linear programming problems.

$2y + 5k < 10$ (i) Maximize $Z = 5x + 12x + 4x$,

$3x + 2y + 4k < 5$ Subject to the constraints

$x, y, k \geq 0$ X, +2x, +x, \$10

2. A manufacture "produces two 2x, —x, +3x, =8

different models: X and Y of the

same product. Model X makes a

contribution of \$50 per unit and

Model Y \$30 per unit towards

total profit. Raw materials r , and

are required for production. At

$X_1, 4X_2, 20$

(ii) Minimize $Z = 15x_1 + 12x_2$,

Subject to the constraints

$2x_1 + 23$

least 18kg of y , and 12kg of 4 , Nae

$2x_1 - 4x_2, \$5$

must be used daily. Also at most >0

34 hours of labour are to be used

utilized. A quantity of 2kg of r , is

(iii) Maximize $Z = 5x_1 + 6x_2$,

Subject to $2x_1 + 4x_2, 24$

$x_1 + 2x_2 = 5, x_1 + 7x_2, 27$

$\rightarrow +5x_1 > 3; x_2, 20$ one

$4x_1 + 7x_2 < 8, 7, 6$. (a) A salesman located in a city A

has decided to travel to city B. He

knows the distances of alternative

routes from city A to city B. He

4. (a) Use the penalty (Big M) method to solve the linear programming problem. The city of origin A is city

programming problem, } 1. The destination city B is city

Minimize $Z = 5x_1 + 3x_2, 10$. Other cities through which the

Subject to the constraints - salesman will have to pass

$2x_1 + 4x_2 < 12$ through are numbered 2 to 9. The

possible routes between cities are in

the following table. The

distance between cities is shown below. The

5. A salesman's problem is to find the

shortest route that covers all the

selected cities from A to B. Solve

(b) Find the optimal solution using the simplex method. Show all

optimal primal basis. Show your workings and put the information

Maximize $Z = 3x_1 + 5x_2$, - In a transition matrix/TPM.

Subject to the constraints $a(1,2) = 4, d(2,5) = 7, d(5,8) = 4, (8,10) = 7,$

$t \$2x_1 + 4, 25, d(2,6) = 10, a(6,8) = 3, 4(6,9) = 7, d(9,10) =$

$1, 43x_1 + x_2 = 2, d(2,7) = 5, d(7,9) = 4, 4(5,9) = 8, d(7,8) = 8,$

ok a $e \neq y, 20, a(1,3) = 6, d(3,5) = 3, d(3,6) = 8, d(3,7) = 4,$

5. Solve the linear programming problem by using the two-phase

simplex method ..

Minimize $Z = x_1 + x_2$,

Subject to the constraints

(b) What is the advantage of the simplex method over the graphical method?

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TIME: 2 HOURS

MTH 219- INTRODUCTION TO OPERATION RESEARCH

INSTRUCTION: ANSWER ANY THREE QUESTIONS

1(a) The ABC printing company is facing a tight financial squeeze and is attempting to cut costs wherever possible. At present it has only one printing contract and, luckily the book is selling well in both the hardcover and paperback editions. It has just received a request to print more copies of this book in either the hardcover or paperback form. The printing cost for hardcover books is #600 per 100 while that for paperback is only #500 per 100.

Although the company is attempting to economize, it does not wish to lay off any employee. Therefore, it feels obliged to run its two printing presses at least 80 and 60 hours per week respectively. Press 1 can produce 100 hardcover books in 2 hours or 100 paperback books in 1 hour. Press 2 can produce 100 hardcover books in 1 hour or 100 paperbacks books in 2 hours. Using the BIG M METHOD, determine how many books of each type should be printed in order to minimize costs.

(b) Use graphical method to solve the following LP Models

Geo A+ GCVOX

Zz

(i) Maximize $Z = 4x + 4y$.

Subject to $x + 2y \leq 10$, $2x + y \leq 72$

$Z \leq 10$

$6x + 6y \leq 36$,

x

$x \leq 6$.

uy 20 ce Cy bee PR _ are

oe soe — aX Ke ov bs ; re c

(ii) Minimize $Z = x - y$. e , -t 074 é

Subject to the constraint $x + y \leq 6$

$-2x + y \leq 8$, Bye Ba ge om ot Moe

$-x + 2y \leq -24$, , 5 ee Os oe TT

$xy \geq 20$; ;

(c) Write the dual of the LP Model,

Maximize $Z = x - y + 3k$,

subject to the constraint

$$x + y + k \leq 10,$$

$$2x - k \leq 2,$$

$$2x - 2y - 3k < 6,$$

$$x, y, k \geq 0$$

2(a) Enumerate the steps involved in carrying out a successful operation research

(ii) Briefly explain what is a system of operation research and its characteristics?

(b) Use graphical method to solve the LP model problem

$$\text{Maximize } Z = 2x + 3y$$

i

gE :

subject to the constraint

$$x + y < 30,$$

$$y \geq 12,$$

$$0 \leq x < 20,$$

$$x - y = 0,$$

$$x, y \geq 0$$

Bai) What are the limitations of Linear Programming? Write short note on the following:

: slack variable, surplus variable, feasible solution, infeasible solution and corner points.

(j) Use the Simplex method to solve the following LP problem

$$\text{Maximize } Z = 3x + 5y + 4k,$$

Subject to the constraints,

$$2x + 3y < 8,$$

$$2y + 5k < 10,$$

$$3x + 2y + 4k < 15,$$

$$x, y, k \geq 0$$

« (ai) Enumerate four special cases in Simplex method

(i) Write short note on Duality

(ii) State the three properties of an LP model.

(>) XYZ Company has 3 departments-Assembly, Painting and Packing, and can make three types of almirahs. An almirah of type 1 requires one hour of assembly, 40 minutes of painting and 20 minutes of packing time respectively. Similarly an almirah of type 11 needs 80 minutes, 20 minutes and one hour respectively. The almirah of type 111 requires 40 minutes each of assembly, painting and packing time. The total time available at assembly, painting and packing department is 600 hours, 400 hours and 800 hours respectively. Determine the number of each type of almirahs that should be produced in order to maximize the profit. The unit profit for types 1, 11, and 111 is \$40, \$80 and \$60 respectively.

' Suppose that the manager of XYZ Company is thinking of renting the production capacities of the three departments to another almirah manufacturer- ABC Company, ABC

(company is interested in minimizing the rental charges. On the other hand he seve
(company would like to know the worth of production hours to ceaanadien Se bape':
(2par:ments to determine the rented rates .

(1) Formulate this problem as an LP problem afd solve it to determine the number of each
t pe of almirahs that should be produced by the XYZ Company in order to maximize its
| rofits.

(>) For LP problem in (a) formulate its dual and ifferpret your results.

Paper 2 of 2 — MTH 219 OPERATION RESEARCH 2022/2023

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN
COLLEGE OF SCIENCE
DEPARTEMENT OF MATHEMATICS
 END OF SEMESTER EXAMINATION FOR MTH 219- (Intro. To Operation Research) 2022/2023

Instruction: Answer Question 1 and any other 2 questions Times: 2 hrs 15minutes

- (a) Using the simplex method only: Maximize $z = 5x + 4y$; Subject to the constraints $6x + 4y \leq 24$, $x + 2y \leq 6$, $-x + y \leq 1$, $y \leq 2$, $x, y \geq 0$. (23 marks).

(b) (i) State and explain the 3 basic properties of a linear programming model. (3 marks)
 (ii) State and explain the 4 features of operation research approach to decision and control problem. (4 marks)
- (a) Using the graphical method to solve the following LP problem

(i) Maximize $z = 2x + y$; Subject to the constraints $5x + y \geq 10$, $x + y \geq 6$, $x + 4y \geq 12$, $x, y \geq 0$. (5 marks)

(ii) Minimize $z = 3x + 2y$; Subject to the constraints $5x + y \geq 10$, $x + y \geq 6$, $x + 4y \geq 12$, $x, y \geq 0$. (5 marks)

(b) A firm makes two products X and Y, and has a total production capacity of 9 tons per day, X and Y requiring the same production capacity. The firm has a permanent contract to supply at least 2 tons of X and at least 3 tons of Y per day to another company. Each ton of X requires 20 machine hours of production time and each ton of Y requires 50 machine hours of production time. The daily maximum possible number of machine hours is 360. All the firms output can be sold, and the profit made is 80 naira per ton of X and 120 naira per ton of Y. determine the production schedule for maximum profit and calculate the profit. (10 marks).
- (a) An advertising agency wishes to reach two types of audiences: customers with annual income great than one target audience A and customers with annual income of less than one target audience B. The total advertising budget is #200,000. One programme of TV advertising cost #50,000; one programme of radio advertising cost #20,000. For contract reason, at least three programmes ought to be on TV and the number of radio programmes must be limited to 5. Surveys indicated that a single reaches 450,000 prospective customers in target audience A and 50,000 in target audience B. one radio programme reaches 20,000 prospective customers in target audience A and 80,000 in target audience B. determine the media mix to maximize the total reach. (10 marks)

(b) (i) State the advantage of two-phase method over other methods. (6 marks)
 (ii) Express the LP model in a dual form. Minimize $z = 3x - 2y + 4m$. Subject to the constraints $3x + 5y + 4m \geq 7$, $6x + y + 3m \geq 4$, $7x - 2y - m \leq 10$, $x - 2y + 5m \geq 3$, $4x + 7y - 2m \geq 2$, $x, y, m \geq 0$ (4 marks)
- (a) Obtain the dual of the LP problem Maximize $z = x - 2y + 3m$. Subject to the constraints $-2x + y + 3m = 2$, $2x + 3y + 4m = 1$, $x, y, m \geq 0$. (2.5 marks)

(b) (i) One unit of product A contributes #7 and requires 3 units of raw material and 2 hours of labour. One unit of product B contributes #5 and requires one unit of raw material and one hour of labour. Availability of raw material at present is 48 units and there are 40 hours of labour. Formulate this problem as a linear programming problem and solve the dual by any method. (2.5 marks)
 (ii) Solve the LP problem by two-phase simplex method. Minimize $z = x + y$ Subject to the constraints $2x + 4y \geq 4$, $x + 7y \geq 7$, $x, y \geq 0$. (15 marks)
- (a) Solve the LP problem by Big-M method. Maximize $z = x + 2y + 3m - t$ Subject to the constraints $x + 2y + 3m = 15$, $2x + y + 5m = 20$, $x + 2y + m + t = 10$, $x, y, m, t \geq 0$. (15 marks)
 (b) Write a short note on the advantages of LP models in modeling real life problems.

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